Flávio Nakasato Cação

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Entrepreneur / AI Researcher / Data Scientist

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Summary

Entrepreneur and AI researcher with ten years of experience building data and AI systems across startups, multinational companies, academic research and social impact organizations. Leads Sumaúma, an AI platform for environmental enforcement in the Amazon rainforest, developed with Imazon, in use by municipal environmental departments in Pará since August 2026. Has built AI tools for climate policy, environmental protection, journalism and finance, for organizations such as Instituto Talanoa, Funbio, World Animal Protection and Brazil's major news outlets. BSc in Physics and MSc in Machine Learning from the University of São Paulo (USP), with peer-reviewed work at ACM CIKM, BRACIS and PROPOR and at climate workshops at ICML and IJCAI-ECAI.

Main Professional Experience

- **Moiru AI / Novelo Data** — São Paulo, SP / Remote
  Founder & Partner — *Mar 2024 / Mar 2022 – Present*
  - **Sumaúma (Project Lead, 2026 – Present)**: AI platform for environmental enforcement in the Amazon rainforest, with Imazon.
  - **AI for the Public Interest (Novelo Data)**: Designs and delivers AI tools for environmental intelligence, policy analysis and media monitoring, used by NGOs, research institutes, newsrooms and public agencies. Political data analysis for nearly all major Brazilian news outlets, including Folha de S.Paulo, Estadão, O Globo and Agência Pública.
- **ADP (Automatic Data Processing, Inc.)** — Remote
  Data Scientist — *Apr 2025 – Jul 2025*
  - **Survey Analysis**: Built a system to identify themes in tens of thousands of employee survey responses.
- **Itaú-Unibanco** — Remote
  Data Scientist — *Mar 2022 – May 2024*
  - **Generative AI Transformation**: Member of the first task force created to bring Generative AI to Brazil's largest bank, designing and promoting its early tools and applications. Developed and deployed generative AI and language processing solutions across several business units.
- **C2D - Data Science Center, University of São Paulo (USP)** — São Paulo, SP / Remote
  Machine Learning Researcher — *Feb 2020 – Jun 2023*
  - **Research**: Language technologies for environmental and ocean data, also within the IBM/USP Center for Artificial Intelligence (C4AI).
  - **Teaching**: Co-taught a graduate mini-course on Deep Reinforcement Learning for students of USP's Graduate Program in Electrical Engineering (PPGEE) and Itaú-Unibanco data scientists (2022).
- **Ada** — São Paulo, SP
  Coding Teacher, Part-time — *Jan 2020 – Dec 2020*
  - **Teaching**: Taught and wrote course materials on Python, databases, cloud computing and neural networks.
- **C6 Bank** — São Paulo, SP
  Senior Data Engineer — *Dec 2019 – Feb 2020*
  - **Data Integration**: Integrated critical databases across cloud platforms under strict security requirements.
- **Cognitivo.AI** — São Paulo, SP
  Data Science Consultant — *Dec 2018 – Aug 2019*
  - **Consulting**: Built data infrastructure and predictive models for client projects.
- **Mira Educação** — São Paulo, SP
  Lead & Senior Data Scientist — *Jan 2017 – Mar 2019*
  - **Data & Impact**: Built the company's data infrastructure and led impact assessment studies as it grew from two pilot schools to nearly one million users.
- **Vá de Táxi, a Porto Seguro startup** — São Paulo, SP
  Business Intelligence Analyst & Data Scientist — *Mar 2016 – Jan 2017*
  - **Data Analysis**: Built performance indicators, mapped fraud patterns and set up a research partnership with academia.

Education

- **Polytechnic School of the University of São Paulo (USP)** — São Paulo, SP
  Master's degree, Computer Engineering (Machine Learning) — *Feb 2020 – Jun 2023*
- **Physics Institute of the University of São Paulo (USP)** — São Paulo, SP
  Bachelor's degree, Physics — *Jan 2011 – Jun 2016*

Selected Projects

- **Sumaúma** – ***Moiru AI / Imazon*** (2026 – Present)
  Project lead of an AI platform to reduce deforestation by strengthening environmental enforcement, built with four municipal environmental departments in Pará (including Altamira and Paragominas) and in use in real enforcement cases since August 2026. Combines an offline field app for inspectors, Imazon's PrevisIA deforestation risk model and AI-drafted technical reports that support fines, embargoes and court cases. Open-source release planned.
- **Climate and Health Alerts App (OCS Fiocruz)** – ***Fiocruz / Brazil's Ministry of Health*** (Apr 2026 – Present, in development)
  Push alerts on climate and health risks (storms, air quality, civil defense) for municipalities, health regions, indigenous lands and quilombola territories, for Fiocruz's Climate and Health Observatory.
- **Anti-Gender Radar** – ***Gênero e Número / JournalismAI / Google News Initiative*** (Feb 2024 – Oct 2024)
  AI lead and product manager of a platform that monitors anti-gender discourse on YouTube, analyzing narratives and actors across thousands of videos daily. The platform received a JournalismAI Innovation Challenge grant (Google News Initiative).
- **Climate Policy Analysis** – ***Instituto Talanoa*** (Mar 2023 – Dec 2024)
  Designed and deployed an AI platform that lets policymakers query 10,000+ climate policy documents in plain language, reducing research time by 60%.
- **Animal Rights Discourse Analysis** – ***World Animal Protection*** (Apr 2024 – Jun 2024)
  Mapped public discourse on animal rights in Brazil at scale to inform campaign design and advocacy.
- **Environmental Inspection Prototypes** – ***with IBAMA's enforcement team*** (2023)
  Built AI prototypes to extract information from field inspection reports and images. The starting point for Sumaúma.
- **Political Discourse Analysis on YouTube** – ***piauí Magazine*** (Jun 2022 – Aug 2022)
  Analyzed 1k+ episodes from a major Brazilian political YouTube channel. Results supported this analysis and this article, among the magazine's most-read pieces of the 2022 presidential election.
- **Environmental News Monitoring** – ***Funbio*** (Jan 2020 – Dec 2021)
  Automated the monitoring of 10k+ news articles a month from 8+ media sources to track environmental threats and narratives.

Selected Peer-Reviewed Publications

- ***Data Intelligence Journal*** **(2024)**: Benchmarks for Pirá 2.0, a Reading Comprehension Dataset about the Ocean, the Brazilian Coast, and Climate Change, Paulo Pirozelli, Marcos M. José, Igor Silveira, Flávio Nakasato, Sarajane M. Peres, et al.
- ***Tackling Climate Change with Machine Learning Workshop*** **(ICML'21)**: *DeepPolicyTracker: Tracking Changes In Environmental Policy In The Brazilian Federal Official Gazette With Deep Learning*, Flávio N. Cação et al.;
- ***Brazilian Conference on Intelligent System*** **(BRACIS'21)**: *DEEPAGÉ: Answering Questions in Portuguese about the Brazilian Environment*, Flávio N. Cação et al.;
- ***International Conference on the Computational Processing of Portuguese*** **(PROPOR'22)**: *Tracking Changes in Environmental Policy in the Brazilian Federal Official Gazette*, Flávio N. Cação et al.
- ***Conference on Information and Knowledge Management*** **(CIKM'21)**: *Pirá: A Bilingual Portuguese-English Dataset for Question-Answering about the Ocean*, André F. A. Paschoal et al.
- ***AI: Modeling Oceans and Climate Change*** **(IJCAI-ECAI'22)**: *The BLue Amazon Brain (BLAB): A Modular Architecture of Services about the Brazilian Maritime Territory*, Paulo Pirozelli et al.

Selected Media

- *Agência Pública* (Sep 2022): co-author of an investigation into how Steve Bannon's *War Room* podcast covered Brazil's 2022 election, based on topic modeling of 923 episodes (link).
- *Folha de S.Paulo* (Oct 2024): quoted on how AI agents work and their limits (link).
- *Folha de S.Paulo* (Aug 2024): quoted as an expert on the originality of an AI model presented as Brazilian (link).
- *Gênero e Número* and *ARTICLE 19 Brazil* (Sep 2025): quoted on the launch of the Anti-Gender Radar (link 1, link 2).

Awards, Programs & Languages

- AmazoniaHack: winner (2023), mentor (2024), organizing team of the 2026 international edition (seven countries).
  Climate Change AI Summer School (Mila, Montreal, 2024), the only participant selected from Brazil in that cohort.
  Youth Climate Leaders (YCL) network, since 2021.
- Portuguese (Native), English (Fluent), Spanish (Intermediate), French (Intermediate)